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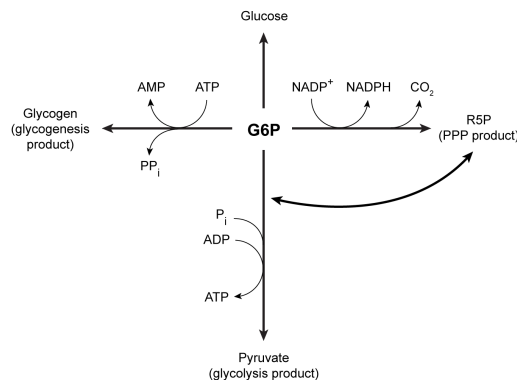


Figure 1 Potential fates of G6P, determined by the needs of the cell

Under low ATP conditions, G6P enters glycolysis, where it is converted to two pyruvate molecules. The energy released in this process produces a net total of two ATP molecules per glucose molecule consumed. The resulting pyruvate molecules can then enter the mitochondria, where they participate in additional metabolic reactions that produce more ATP.

When the cell experiences oxidative stress, G6P enters the pentose phosphate pathway (PPP) and produces NADPH, a reducing agent. During this process, G6P loses one carbon atom in the form of CO_2 and is converted to ribose 5-phosphate (R5P), which is required for nucleotide synthesis. In addition to low NADPH conditions, G6P may also enter the PPP during replication, when nucleotides are needed for DNA synthesis. If nucleotides are not needed but energy is, R5P can be converted back to glycolytic intermediates and reenter glycolysis. Each glucose molecule that is converted to R5P prior to glycolysis yields an average of 1.67 ATP molecules.

When all the cell's metabolic requirements are met, additional G6P can enter glycogenesis and produce glycogen, which stores glucose for later use; this process consumes ATP. When signal molecules bind G protein–coupled receptors on muscle cells, glycogenolysis is activated to release individual monomers from glycogen, allowing them to enter the appropriate pathway.

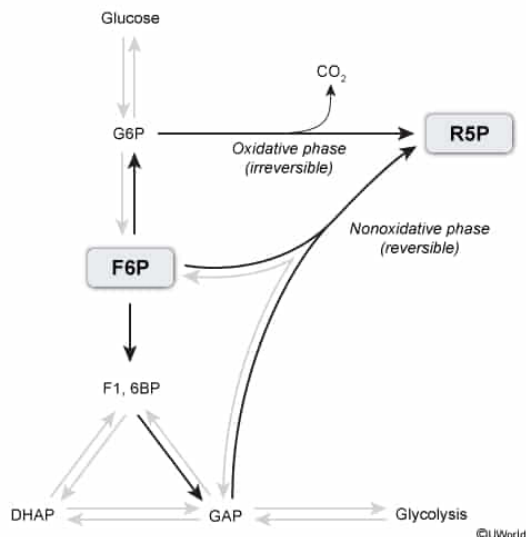
A test tube contains ATP, NADP^+ , and all the enzymes of glycolysis and the pentose phosphate pathway in a buffered solution. In this system, which of the following molecules could be metabolized to R5P, assuming no other intermediate is added?

- ☒ A. Fructose 6-phosphate [66%]
- ☐ B. Fructose 1,6-bisphosphate [9%]
- ☐ C. Glyceraldehyde 3-phosphate [16%]
- ☐ D. Dihydroxyacetone phosphate [6%]

Correct

66% answered correctly

Explanation:



Ribose 5-phosphate (R5P) is a phosphorylated five-carbon sugar (pentose phosphate) produced in the **pentose phosphate pathway (PPP)**. The PPP consists of two major parts: an **irreversible oxidative phase** and a **reversible nonoxidative phase**. Fructose 6-phosphate (F6P) can enter either phase by different means.

- F6P is readily converted to G6P by the glycolysis enzyme phosphoglucose isomerase. G6P can then enter the oxidative phase of the PPP, where it is **decarboxylated** (loss of CO₂) and converted to R5P.
- The nonoxidative phase recycles R5P by combining three R5P molecules and **rearranging their carbon atoms** to produce two F6P and one glyceraldehyde 3-phosphate (GAP). Because this phase is reversible, two F6P and one GAP can also combine and rearrange to become three molecules of R5P. After some of the F6P in the system has been converted to GAP molecules by glycolysis enzymes, the resulting GAP can then combine with the remaining F6P. Therefore, F6P can also enter the PPP through the nonoxidative phase to produce R5P.

(Choice B) Fructose 1,6-bisphosphate (F1,6BP) must be converted to F6P before it can enter the PPP. This requires the **gluconeogenic** enzyme fructose 1,6-bisphosphatase, but this enzyme is not part of glycolysis and is therefore absent from the test tube.

(Choices C and D) GAP can only enter the PPP by *combining with F6P*. If GAP alone were added to the test tube, some of it could be converted to dihydroxyacetone phosphate (DHAP), and the remaining GAP and DHAP could combine to form F1,6BP. However, the test tube lacks enzymes of gluconeogenesis, including fructose-1,6-bisphosphatase, which is necessary to convert F1,6BP to F6P. Therefore, neither GAP nor DHAP *alone* is sufficient for R5P synthesis in this system.

Educational objective:

The pentose phosphate pathway (PPP) can produce ribose 5-phosphate by decarboxylation of glucose 6-phosphate (oxidative phase) or by rearranging the carbon atoms in fructose 6-phosphate (F6P) and glyceraldehyde 3-phosphate (nonoxidative phase). F6P can be converted to all these intermediates by the enzymes of glycolysis and the PPP.

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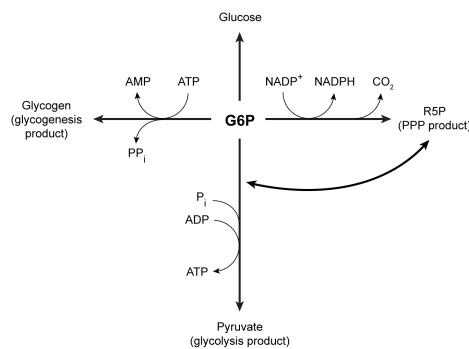


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When all the cell's metabolic requirements are met, additional G6P can enter glycogenesis and produce glycogen, which stores glucose for later use; this process consumes ATP. When signal molecules bind G protein–coupled receptors on muscle cells, glycogenolysis is activated to release individual monomers from glycogen, allowing them to enter the appropriate pathway.

Based on the information in the passage, if glucose enters the pentose phosphate pathway prior to glycolysis, what is the energy loss compared to glucose that enters glycolysis directly? (Note: The free energy of ATP hydrolysis is –30 kJ/mol.)

- ☐ A. 15 kJ/mol [13%]
- ☐ B. 0 kJ/mol [8%]
- ☒ C. –10 kJ/mol [62%]
- ☐ D. –20 kJ/mol [14%]

Correct
63% answered correctly

Explanation:

	Direct glycolysis	Glycolysis after PPP
ATP yield per glucose	2	1.67
Energy of ATP produced (ATP yield × –30 kJ/mol)	–60 kJ/mol	–50 kJ/mol
Energy difference: –60 kJ/mol – (–50 kJ/mol) = –10 kJ/mol		

The passage states that when glucose enters **glycolysis** directly, **2 ATP molecules** are produced per glucose molecule consumed. It further states that if glucose is first diverted through the **PPP**, on average **only 1.67 ATP molecules** are produced per glucose molecule. This reduction occurs because a carbon atom is lost as CO_2 in the PPP, reducing the total number of pyruvate molecules that can be produced during glycolysis and consequently reducing the net ATP synthesized in the process.

The free energy available in each ATP molecule is -30 kJ/mol , so the 2 ATP molecules produced per glucose molecule carry -60 kJ/mol . The 1.67 ATP molecules produced by glycolysis after the PPP yield -50 kJ/mol . The energy cost of the PPP can be calculated as the **difference in the free energies** of each path: $-60 \text{ kJ/mol} - (-50 \text{ kJ/mol}) = -10 \text{ kJ/mol}$.

Alternatively, the PPP may be considered as effectively reducing the yield by one third of an ATP molecule per glucose molecule. Therefore, one third of the -30 kJ/mol available in ATP (ie, -10 kJ/mol) is lost by diversion into the PPP.

(Choice A) 15 kJ/mol is a positive number. when *energy leaves the system* (rather than being stored), it must be reflected as a *negative value*.

(Choice B) Energy is lost from the system when glucose enters the PPP, so 0 kJ/mol cannot be the correct answer.

(Choice D) The magnitude of the energy cost would be 20 kJ/mol if direct glycolysis were to produce one ATP molecule per glucose molecule.

Educational objective:

Glycolysis produces energy in the form of ATP, some of which may be sacrificed by pursuing other pathways that satisfy other needs. The energy cost of these pathways can be calculated as the difference in free energy available in the ATP produced by each pathway. A negative free energy difference between pathways indicates that energy was lost and is manifested as fewer ATP molecules produced.

Time spent: 0 sec
Subject:
Biochemistry

QID: 400993
System:
Metabolic Reactions

Upon entering cells, glucose is quickly phosphorylated by hexokinase to produce glucose 6-phosphate (G6P). This phosphorylation event prevents glucose from exiting the cell. G6P may then enter one of several metabolic pathways, depending on the needs of the cell (Figure 1).

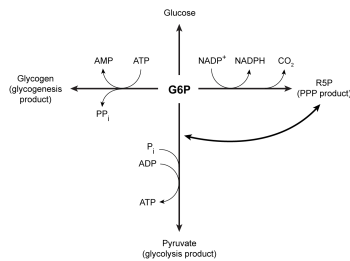


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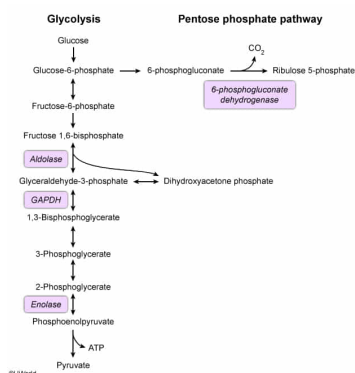
When all the cell's metabolic requirements are met, additional G6P can enter glycogenesis and produce glycogen, which stores glucose for later use; this process consumes ATP. When signal molecules bind G protein–coupled receptors on muscle cells, glycogenolysis is activated to release individual monomers from glycogen, allowing them to enter the appropriate pathway.

Under low ATP conditions, which of the following enzymes would be LEAST beneficial to the cell?

- ☐ A. Aldolase [7%]
- ☐ B. Enolase [8%]
- ☐ C. Glyceraldehyde 3-phosphate dehydrogenase [10%]
- ☒ D. 6-Phosphogluconate dehydrogenase [72%]

Correct
73% answered correctly

Explanation:



As stated in the passage, under low ATP conditions the G6P in a cell will enter **glycolysis**, which **produces ATP**. The final ATP molecules of this pathway are produced by the final enzyme in the pathway, pyruvate kinase. Therefore, all **glycolysis enzymes** upstream of pyruvate kinase are **required for maximum ATP** synthesis, as their absence would prohibit pyruvate kinase function. These enzymes include the following:

- Aldolase catalyzes the cleavage of fructose 1,6-bisphosphate into glyceraldehyde 3-phosphate (GAP) and dihydroxyacetone phosphate (**Choice A**).
- Enolase catalyzes the conversion of 2-phosphoglycerate to phosphoenolpyruvate and is the second-to-last step in the pathway (**Choice B**).
- Glyceraldehyde 3-phosphate dehydrogenase, also known as GAPDH, catalyzes the conversion of GAP to 1,3-bisphosphoglycerate in a step that immediately precedes the first ATP-synthesis step of glycolysis (**Choice C**).

6-Phosphogluconate dehydrogenase (6PGD) is the enzyme in the **pentose phosphate pathway** that catalyzes the conversion of 6-phosphogluconate to ribulose-5-phosphate (Ru5P). It is in this step that a carbon atom is converted to CO_2 , decreasing the overall potential ATP output. Therefore, **6PGD** would be **less beneficial** to a cell deficient in ATP than would any glycolytic enzyme.

Educational objective:

Of the metabolic pathways that can be taken by glucose, glycolysis and its downstream pathways produce the most ATP. The enzymes of glycolysis are of the most benefit to cells lacking ATP whereas enzymes of other pathways are beneficial for satisfying other needs.

Time spent:0 sec**Subject:**

Biochemistry

QID:400994**System:**

Metabolic Reactions

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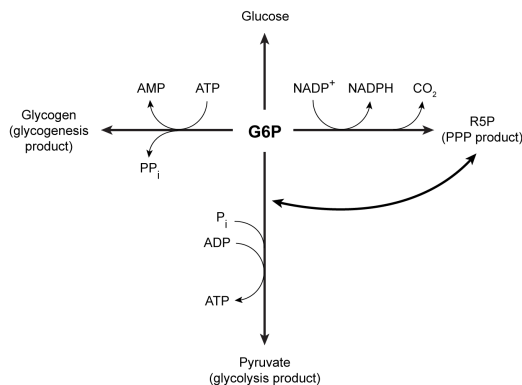


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When the cell experiences oxidative stress, G6P enters the pentose phosphate pathway (PPP) and produces NADPH, a reducing agent. During this process, G6P loses one carbon atom in the form of CO₂ and is converted to ribose 5-phosphate (R5P), which is required for nucleotide synthesis. In addition to low NADPH conditions, G6P may also enter the PPP during replication, when nucleotides are needed for DNA synthesis. If nucleotides are not needed but energy is, R5P can be converted back to glycolytic intermediates and reenter glycolysis. Each glucose molecule that is converted to R5P prior to glycolysis yields an average of 1.67 ATP molecules.

When all the cell's metabolic requirements are met, additional G6P can enter glycogenesis and produce glycogen, which stores glucose for later use; this process consumes ATP. When signal molecules bind G protein–coupled receptors on muscle cells, glycogenolysis is activated to release individual monomers from glycogen, allowing them to enter the appropriate pathway.

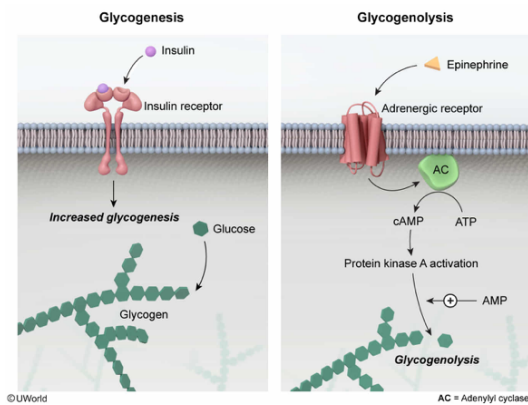
The fact that glycogenolysis in muscle cells is activated by signal molecules binding to G protein–coupled receptors suggests that glycogenolysis can be induced by which of the following?

- ☒ A. Activation of adenylate cyclase [79%]
- ☐ B. Inhibition of glycogen phosphorylase [4%]
- ☐ C. Increased blood glucose concentration [5%]
- ☐ D. Depletion of cytosolic cAMP [11%]

Correct

79% answered correctly

Explanation:



Intracellular glycogen levels are controlled by two opposing pathways, **glycogenesis** (synthesis) and **glycogenolysis** (degradation). In muscle cells, glycogenesis occurs when sufficient blood glucose induces insulin synthesis, and glycogenolysis is induced when hormones such as epinephrine bind G protein–coupled receptors and activate an intracellular signal cascade.

Upon epinephrine binding, G protein–coupled receptors activate the enzyme **adenylate cyclase**, which converts ATP into cyclic AMP (cAMP) in the cytosol. Newly synthesized cAMP then **allosterically activates protein kinase A**, initiating a phosphorylation cascade that ultimately activates **glycogen phosphorylase**, the first enzyme in glycogenolysis.

(Choice B) Because glycogen phosphorylase is the first enzyme in the glycogenolysis pathway, inhibiting it would lead to *decreased* glycogenolysis.

(Choice C) Increased blood glucose levels lead to increased insulin expression. Insulin induces *glycogenesis* (synthesis of glycogen), the opposite of glycogenolysis.

(Choice D) Cytosolic cAMP is required for activation of protein kinase A and, ultimately, glycogen phosphorylase. Depletion of cAMP would lead to decreased glycogenolysis.

Educational objective:

Glycogenesis and glycogenolysis are opposing pathways used for storing glucose and for using stored glucose, respectively. Glycogenesis is induced by insulin in response to high blood glucose levels. In muscle cells, glycogenolysis is induced by epinephrine binding to G protein–coupled receptors, which ultimately activates glycogen phosphorylase.

Time spent: 0 sec

Subject:
Biochemistry

QID: 400995

System:
Metabolic Reactions

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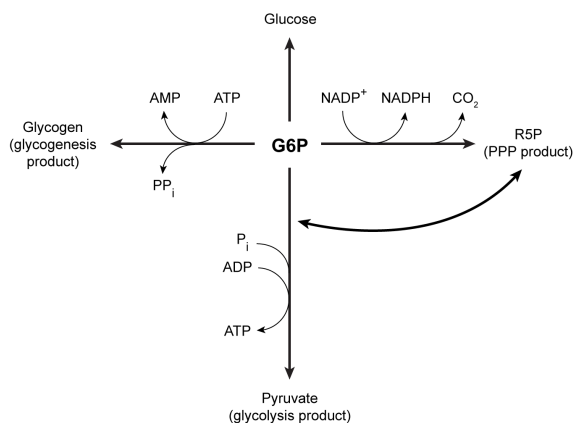


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The glycogen branching enzyme forms a glycosidic bond between carbon 1 of one glucose unit and which of the following?

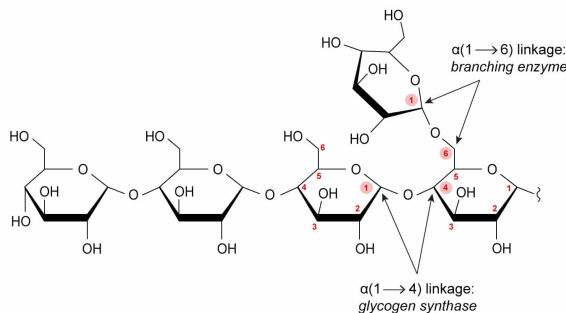
- ☐ A. Carbon 3 on a glucose unit in the growing glycogen chain [3%]
- ☐ B. Carbon 4 on a glucose unit in the growing glycogen chain [25%]

- ☐ C. Carbon 5 on a glucose unit in the growing glycogen chain [2%]
- ✓ ☒ D. Carbon 6 on a glucose unit in the growing glycogen chain [68%]

Correct

68% answered correctly

Explanation:



Glycogen is composed of several **linked glucose** molecules. The majority of these units are connected to each other in an **$\alpha(1 \rightarrow 4)$ linkage**, meaning carbon 1 of each incoming glucose molecule becomes linked in the **α conformation** to carbon 4 of the final carbon atom in the growing chain. This reaction is catalyzed by **glycogen synthase**.

However, glycogen is a **branched molecule** rather than linear. The **glycogen branching enzyme** creates these branches by catalyzing the formation of **$\alpha(1 \rightarrow 6)$ linkages**, acting on approximately every tenth glucose unit in the chain. In this case, carbon 1 of the incoming glucose molecule in the α conformation is linked to carbon 6 of a glucose molecule in the growing glycogen chain. This branch then begins a new series of $\alpha(1 \rightarrow 4)$ linkages.

(Choices A and C) Carbon 3 and carbon 5 are not involved in glycosidic bonds in glycogen.

(Choice B) Carbon 4 is involved in glycosidic bonds in glycogen, but glycogen synthase catalyzes the formation of these bonds, not the branching enzyme.

Educational objective:

Glycogen is a branched molecule composed of several linked glucose units. Each incoming unit is linked to the growing chain by a bond between carbon 1 of the incoming glucose and either carbon 4 or carbon 6 of a glucose unit in the growing glycogen chain. $\alpha(1 \rightarrow 4)$ linkages are facilitated by glycogen synthase whereas $\alpha(1 \rightarrow 6)$ linkages are facilitated by the glycogen branching enzyme.

Time spent: 0 sec

Subject:
Biochemistry

QID: 400996

System:
Metabolic Reactions

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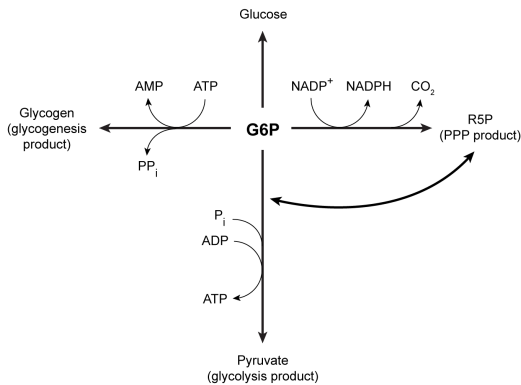


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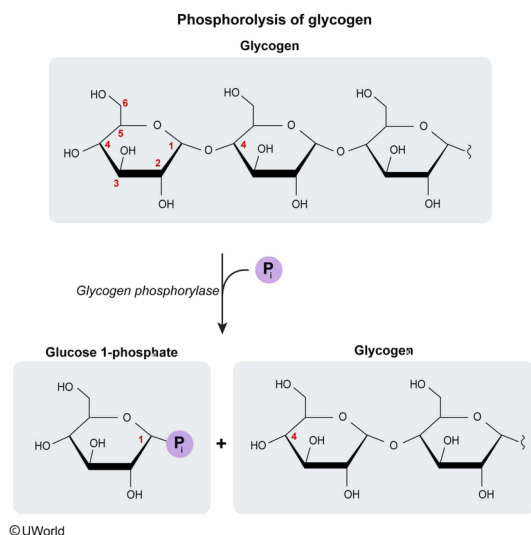
When all the cell's metabolic requirements are met, additional G6P can enter glycogenesis and produce glycogen, which stores glucose for later use; this process consumes ATP. When signal molecules bind G protein–coupled receptors on muscle cells, glycogenolysis is activated to release individual monomers from glycogen, allowing them to enter the appropriate pathway.

When the cell needs glucose, glycogenolysis is upregulated, beginning with the activation of glycogen phosphorylase. Glycogen phosphorylase catalyzes the production of which of the following molecules?

- ☐ A. UDP-glucose [15%]
- ☐ B. UDP-galactose [2%]
- ✔ ☒ C. Glucose 1-phosphate [55%]
- ☐ D. Glucose 6-phosphate [26%]

Correct

56% answered correctly



Glycogen phosphorylase breaks the bond between two adjacent glucose molecules in a glycogen chain by **phosphorolysis**, or the addition of a phosphate molecule across a bond. The bond broken by this reaction is an $\alpha(1 \rightarrow 4)$ linkage, so the phosphate must be added to either carbon 1 or carbon 4 of a glucose unit. The glucose unit that is released was attached by carbon 1, and it is to this carbon atom that phosphate is added, producing **glucose 1-phosphate (G1P)**.

(Choice A) UDP-glucose is produced by the combination of G1P with UTP along with the release of pyrophosphate. This reaction, catalyzed by UDP-glucose pyrophosphorylase, is involved in glycogenesis but not glycogenolysis.

(Choice B) UDP-galactose is an intermediate of galactose metabolism. It is converted to UDP-glucose and later G1P but is not itself a product of glycogenolysis.

(Choice D) Glucose 6-phosphate (G6P) is produced during glycogenolysis when phosphoglucomutase moves the phosphate group from carbon 1 to carbon 6. It is not produced by glycogen phosphorylase.

Educational objective:

Glycogen is degraded by phosphorolysis, or the addition of phosphate to break a bond, to produce glucose 1-phosphate (G1P). G1P is subsequently transformed into glucose 6-phosphate for use in glycolysis or the pentose phosphate pathway.

Time spent: 0 sec
Subject:
 Biochemistry

QID: 400997
System:
 Metabolic Reactions